That is an excellent question. The "correlation" mentioned in the video is a *crucial* concept, and it's easy to miss.

The video is **not** talking about the correlation between your *input variable* (like `Weight`) and your *output variable* (like `Size`).

It's talking about correlation **between two or more *input* variables**. This is a problem called **multicollinearity**.

---

## What are Correlated Variables?

Imagine you are trying to predict `Size` using these three input variables:

1. **`Weight_in_kg`**
2. **`Weight_in_pounds`**
3. **`BMI`**

`Weight_in_kg` and `Weight_in_pounds` are almost perfectly correlated (they measure the same thing). `Weight_in_kg` and `BMI` are also highly correlated.

This creates a huge problem for a regression model:

- When the model sees a change in `Size`, it can't tell if it should give credit to `Weight_in_kg` or `Weight_in_pounds`.
- The model becomes unstable. It might give a huge positive slope to `Weight_in_kg` and a huge negative slope to `Weight_in_pounds`, which technically "cancels out" but is meaningless.

---

## How Each Regression Model Handles This Problem

This is the most important part of choosing between the three models.

#### 1. Lasso (L1) Regression

- Lasso will arbitrarily pick **one** of the correlated variables and set the others to **zero**.
- It might pick `Weight_in_kg` and completely delete `Weight_in_pounds` and `BMI` from the model.
- **The Problem:** This is very unstable. If you change your data slightly, it might pick `BMI` instead and delete the other two. Your model will be completely different every time.

#### 2. Ridge (L2) Regression

- Ridge will **shrink all** the correlated variables *together*.

- It will keep `Weight_in_kg`, `Weight_in_pounds`, and `BMI` in the model, but it will give them all smaller, similar-sized slopes.
- **The Benefit:** This is much more stable than Lasso's "all-or-nothing" approach.

#### 3. Elastic Net (The Hybrid)

- This is why Elastic Net is so powerful. It combines the L1 and L2 penalties.
- The L2 (Ridge) part **groups** the correlated variables together, just like Ridge does.
- The L1 (Lasso) part then decides whether to **keep that entire group** or **set the entire group to zero**.

This "grouping" behavior is the main advantage. Elastic Net doesn't just pick one winner like Lasso; it identifies that `Weight`, `BMI`, and `Weight_in_pounds` are all one "concept" and treats them as a single block.